(Please write your Enrollment Number) Enrollment...

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: IT/MAE/DMAM, Semester:2
(July, 2023)

Subject Code: BCS 110	Subject: Programming in C Language
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8= 20)
	(a)	Explain registers in terms of computer system.	
	(b)	What is the essence of using algorithm and flowchart in programming language?	
	(c)	What is the difference between #include<...> and #include"..."?	

	(d)	Consider the following program:
		int main()
		{
		float a;
		printf("Enter a:");
		scanf("%f",&a);
		switch(a)
		{
		case 1: printf("JK");
		break;
		case 2: printf("Jenny");
		break;
		default: printf("Bye");
		}
		return 0;
		}
		What is the output of the above program?	

	(e)	Is the name of the formal arguments or parameters in a function prototype mandatory? Justify your answer with example.	

	(f)	Consider the following program:
		void main()
		{
		int i,j;
		for(i=1,j=0;i<=5,j<3;i++)
		{
		printf("%d %d",i,j);
		}
		}	

	(g)	What is difference between the following declarations? int(*a) []; int *a[];	
	(h)	What is designated initialization of structure? Explain with example.	

UNIT-I

Q2	a)	Explain different types of finite storage class in C with example.	(5)
	b)	Write short notes on: i) Batch Operating System ii) Multiprogramming Operating System iii) Time Sharing Operating System.	(5)

Q3	a) Differentiate between the Random Access Memory and Cache Memory. How does cache memory work? b) Draw a flowchart and write an algorithm to find the sum of digits of a number n.	(5) (5)	
	UNIT-II		
Q4	a) Explain different data types in C with example. b) How does the break statement differ from continue statement? Support your answer with suitable example.	(5) (5)	
Q5	a) Explain logical operators in C with example. b) WAP in C to swap two numbers without using a third variable.	(5) (5)	
	UNIT-III		
Q6	a) Is the working logic of while loop same as do-while loop? Justify your answer with suitable example. b) WAP in C to print the following pattern as: 1 23 456 78910 1112131415	(5) (5)	
Q7	a) Explain tail and non-tail recursive function with example. b) WAP in C to print the transpose of a matrix.	(5) (5)	
	UNIT-IV		
Q8	a) What is array of pointers? Support your answer with example. b) Differentiate between Call by Value and Call by Reference with suitable example.	(5) (5)	
Q9	a) How does the structure differ from union in C? Take example and support your answer. b) Can we pass array as an argument to a function? Support your answer with an example.	(5) (5)	